Deepthi kakani

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Summary

Aspiring B.Tech Artificial Intelligence student with strong foundations in Data Structures, Algorithms, and Artificial Intelligence concepts. Skilled in C, Python, Java, and problem-solving with experience in developing AI-based applications and intelligent systems. Passionate about building impactful real-world solutions through innovation, teamwork, and continuous learning.

Education

Anurag University, Hyderabad — **Aug. 2024 – Present**
Bachelor of Technology in Artificial Intelligence (CGPA : 8.85) — *Hyderabad, Telangana*

Sri Chaitanya Jr. College, Khammam — **2022 – 2024**
Class XII, Telangana State Board (Percentage: 97.8%) — *Khammam, Telangana*

Shine High School, Bonakal — **2021 – 2022**
Class X, Telangana State Board (CGPA: 9.8) — *Bonakal, Telangana*

Technical Skills

Programming Languages: C, Python, Java, SQL
Databases: MySQL, MongoDB
Core Concepts: Data Structures, Algorithms, DBMS
Tools & Platforms: VS Code, Jupyter Notebook

Certifications

- Problem Solving - HackerRank
- AVINYA 2K25 National Level Hackathon - Participation
- Programming fundamentals in Python - Infosys Springboard
- Database Management System - Infosys Springboard
- Introduction to MongoDB - MongoDB University

Projects

AI-Powered Coordinated Traffic Management System

- Developed an AI-based traffic management system using Python and Flask, with a user interface built in HTML, to predict traffic and control signals.
- Implemented a Linear Regression model using Scikit-learn and NumPy to predict traffic volume based on time and junction data.
- Designed a multi-intersection coordination system using NetworkX and Dijkstra's algorithm to optimize traffic flow.
- Created dynamic signal control logic to adjust green signal timing based on traffic density.
- Implemented an emergency vehicle priority feature (green corridor) to allow faster movement of ambulances.
- Visualized the traffic network using Matplotlib and demonstrated a simulated 30–40% reduction in travel time.

AI Research Assistant Agent

- Developed an AI Research Assistant Agent using Python to summarize and analyze research papers automatically.
- Built a simple user interface using HTML, CSS, and JavaScript for uploading papers and viewing summaries.).
- Implemented text summarization and keyword extraction using basic NLP concepts in Python.
- Integrated online research paper sources to collect scientific articles for analysis. Used Flask for backend development and connecting the AI functionalities with the frontend.
- Improved research efficiency by reducing manual literature review time through AI automation.

Sudoku Solver using Graphs

- Developed a Sudoku solver using Python with a graph-based approach and Depth First Search (DFS).
- Represented Sudoku cells as nodes and constraints as edges.
- Implemented recursive backtracking logic in VS Code to generate valid solutions.
- Used VisuAlgo for visualizing DFS traversal and debugging.
- Optimized the program to improve time and memory efficiency.